

Antarctica

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Antarctica is, on average, the coldest, driest, and windiest of the continents, and it has the highest average elevation. It is mainly a polar desert, with annual precipitation of over 200 mm (8 in) along the coast and far less inland. About 70% of the world's freshwater reserves are frozen in Antarctica, which, if melted, would raise global sea levels by almost 60 metres (200 ft). Antarctica holds the record for the lowest measured temperature on Earth, −89.2 °C (−128.6 °F). The coastal regions can reach temperatures over 10 °C (50 °F) in summer. Native species of animals include mites, nematodes, penguins, seals and tardigrades. Where vegetation occurs, it is mostly in the form of lichen or moss.

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Area	14,200,000 km ² 5,500,000 sq mi^[1]
Population	1,000 to 5,000 (seasonal)
Population density	<0.01/km ² <0.03/sq mi
Internet TLD	.aq
Largest settlements	McMurdo Station Other research stations
UN M49 code	010

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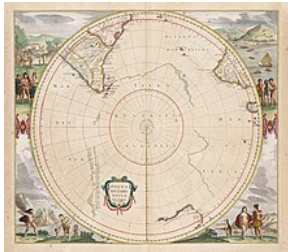
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Composite satellite image of Antarctica (2002)



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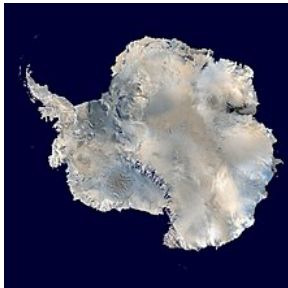
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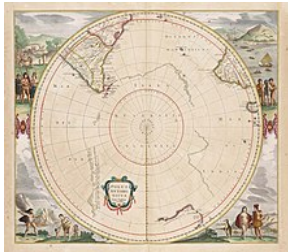
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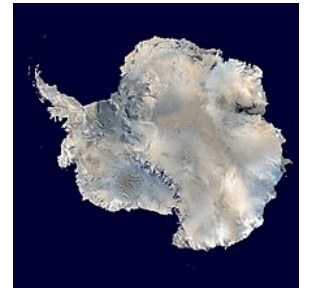
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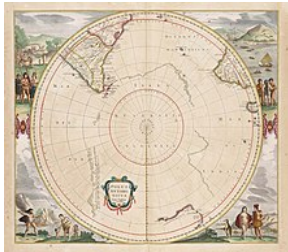
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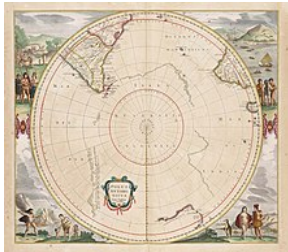
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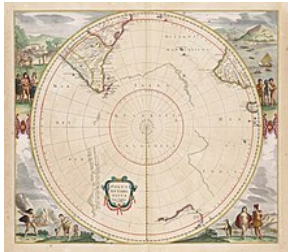
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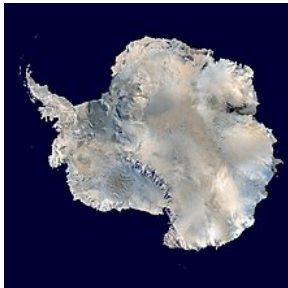
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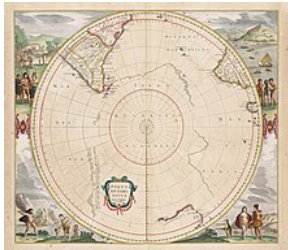
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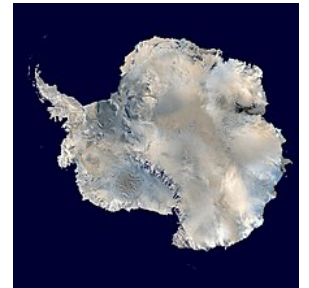
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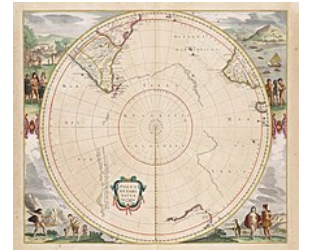
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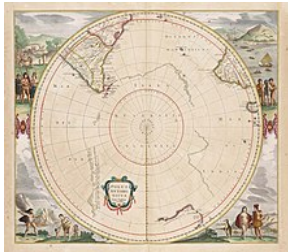
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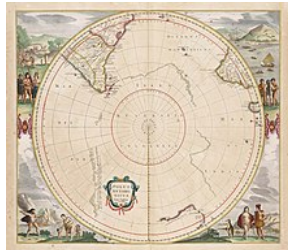
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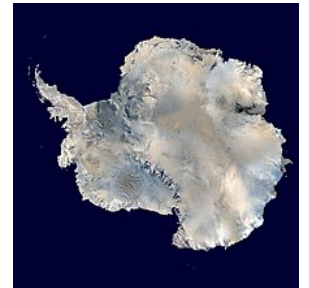
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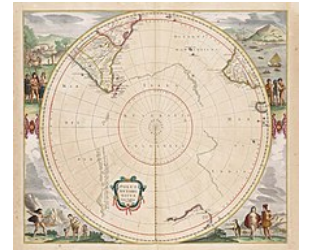
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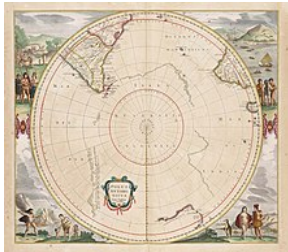
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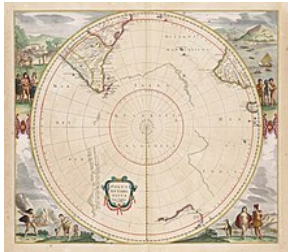
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A speculative representation of Antarctica labelled as 'Terra Australis Incognita' on Jan Janssonius's *Zeekaart van het Zuidpoolgebied* (1657), Het Scheepvaartmuseum

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From the end of the Neoproterozoic era to the Cretaceous, Antarctica was part of the supercontinent Gondwana.^[28] Modern Antarctica was formed as Gondwana gradually broke apart beginning around 183 Ma.^[29] For a large proportion of the Phanerozoic, Antarctica had a tropical or temperate climate, and it was covered in forests.^[30]

Palaeozoic era (540–250 Ma)



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Glossopteris sp. leaf from the Permian of Antarctica

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Africa separated from Antarctica in the Jurassic around 160 Ma, followed by the Indian subcontinent in the early Cretaceous (about 125 Ma).^[47] During the early Paleogene, Antarctica remained connected to South America via the Isthmus of Scotia as well as to southeastern Australia. Fauna from the La Meseta Formation in the Antarctic Peninsula, dating to the Eocene, is very similar to equivalent South American faunas; with marsupials, xenarthrans, lioptern, and astrapotherian ungulates, as well as gondwanatheres and possibly meridiolestidans.^{[48][49]} Marsupials are thought to have dispersed into Australia via Antarctica by the early Eocene.^[50]

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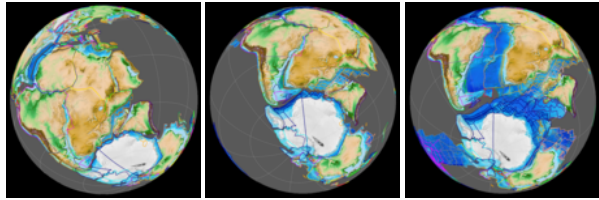
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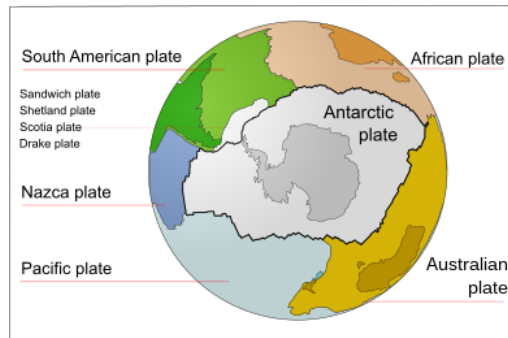


Breakup of Gondwana at c. 150 Ma (left), c. 126 Ma (centre) and at c. 83 Ma (right)^[note 3]

Present day

The geology of Antarctica, largely obscured by the continental ice sheet,^[55] is being revealed by techniques such as remote sensing, ground-penetrating radar, and satellite imagery.^[56] Geologically, West Antarctica closely resembles the South American Andes.^[57] The Antarctic Peninsula was formed by geologic uplift and the transformation of sea bed sediments into metamorphic rocks.^[58]

West Antarctica was formed by the merging of several continental plates, which created a number of mountain ranges in the region, the most prominent being the Ellsworth Mountains. The presence of the West Antarctic Rift System has resulted in volcanism along the border between West and East Antarctica, as well as the creation of the Transantarctic Mountains.^[59]



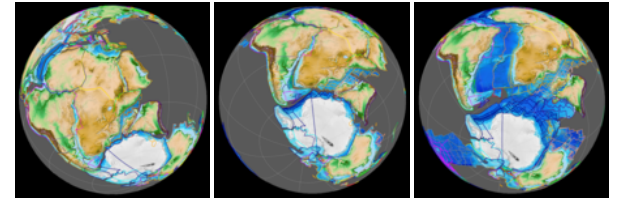
The Antarctic Plate

East Antarctica is geologically varied. Its formation began during the Archean Eon (4,000 Ma–2,500 Ma), and stopped during the Cambrian Period.^[60] It is built on a craton of rock, which is the basis of the Precambrian Shield.^[61] On top of the base are coal and sandstones, limestones, and shales that were laid down during the Devonian and Jurassic periods to form the Transantarctic Mountains.^[62] In coastal areas such as the Shackleton Range and Victoria Land, some faulting has occurred.^{[63][64]}

Coal was first recorded in Antarctica near the Beardmore Glacier by Frank Wild on the Nimrod Expedition in 1907, and low-grade coal is known to exist across many parts of the Transantarctic Mountains.^[65] The Prince Charles Mountains contain deposits of iron ore.^[66] There are oil and natural gas fields in the Ross

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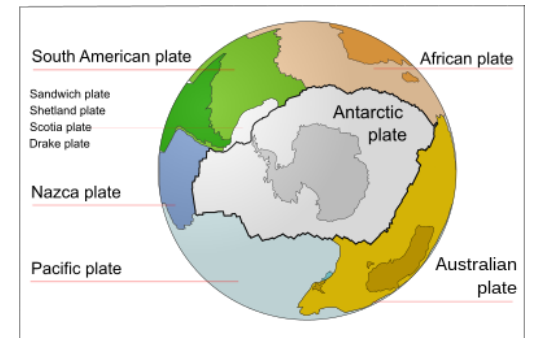


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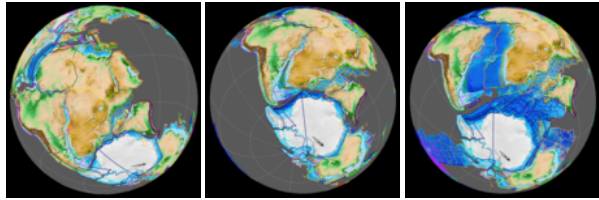
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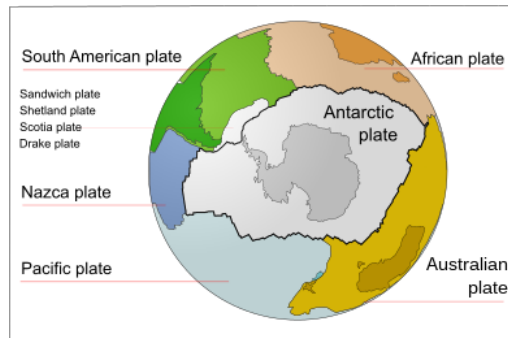
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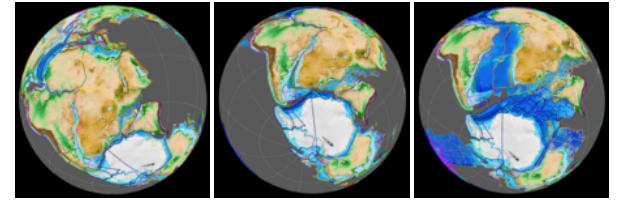
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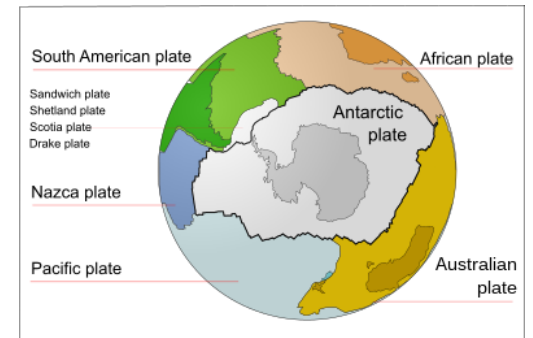
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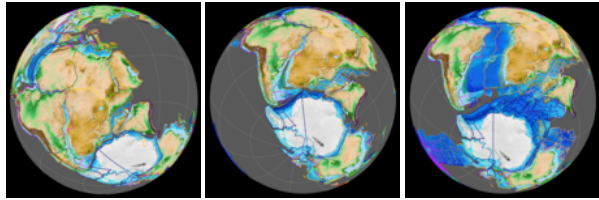
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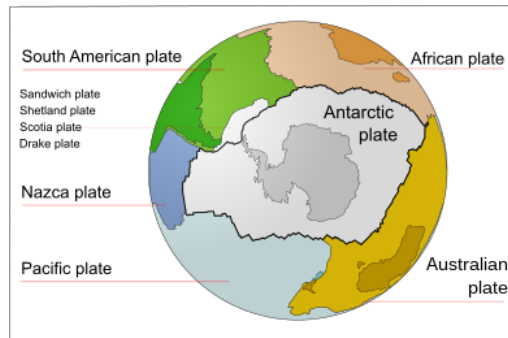


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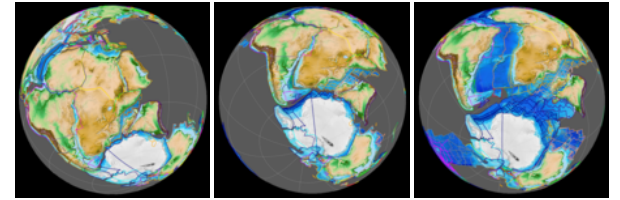
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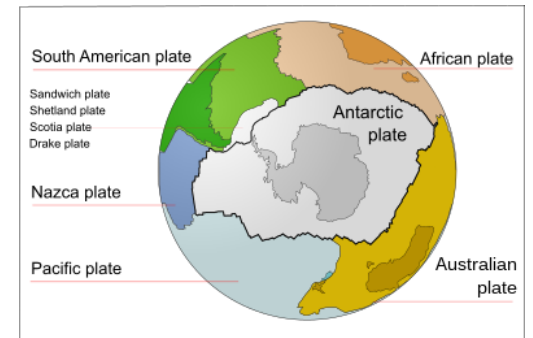


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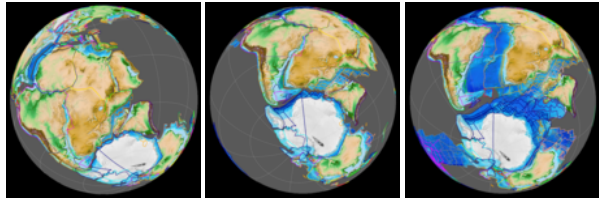
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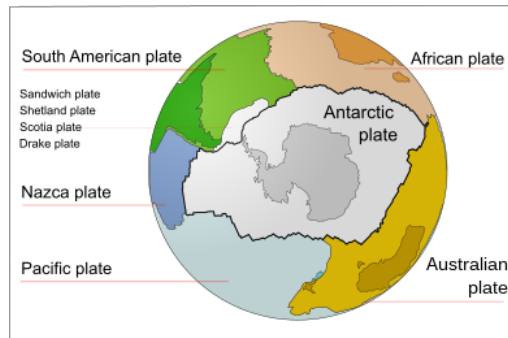
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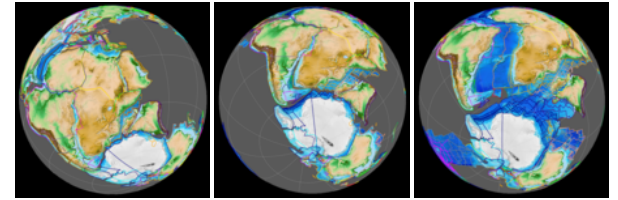
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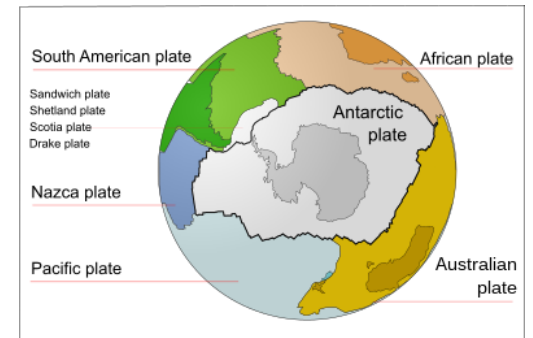
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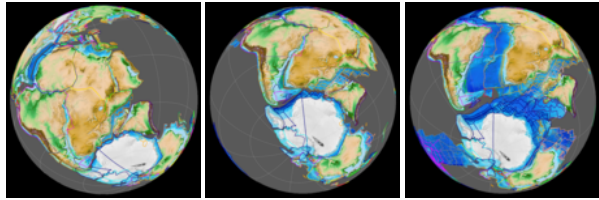
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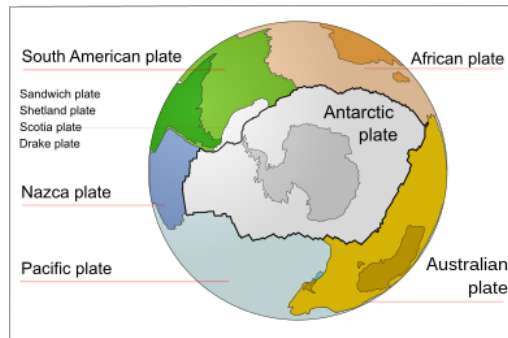
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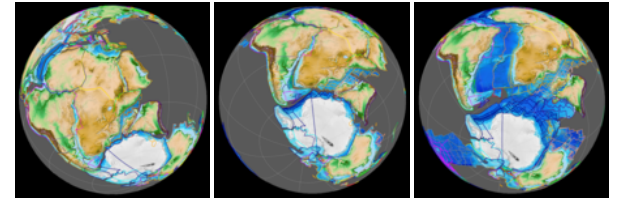
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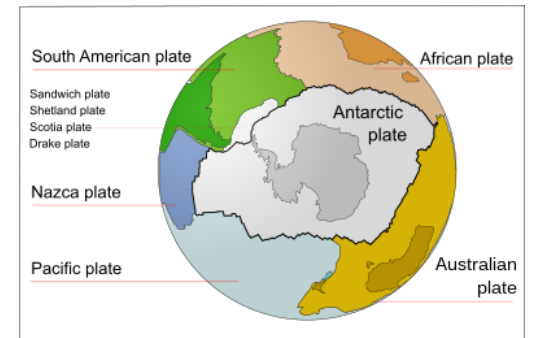
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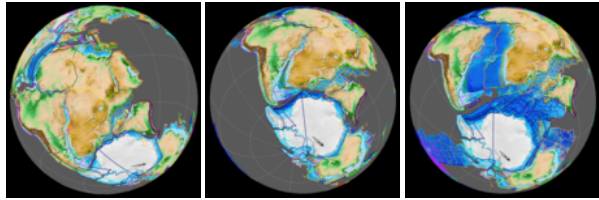
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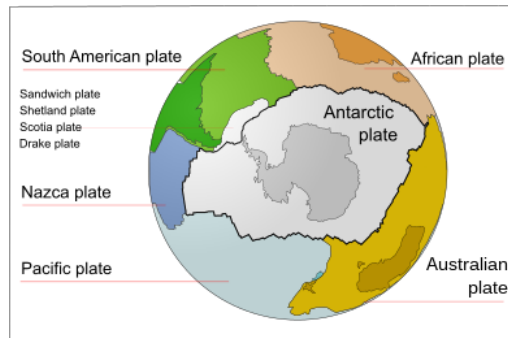


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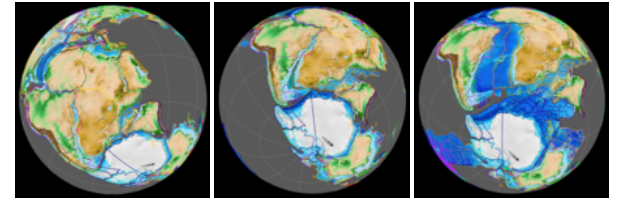
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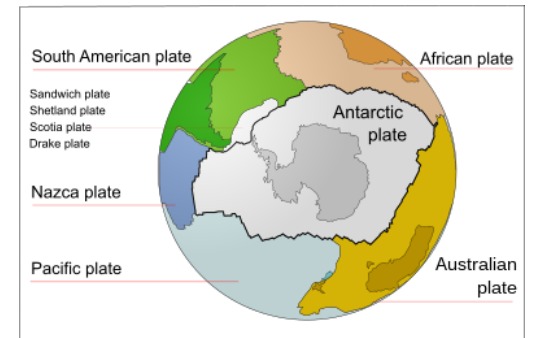


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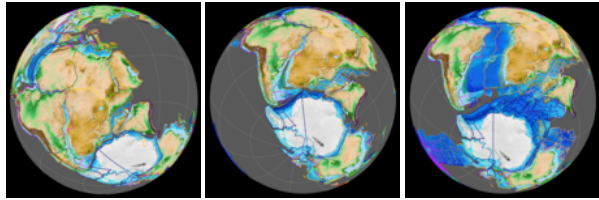
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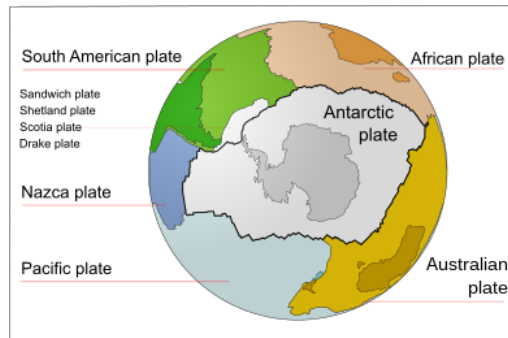


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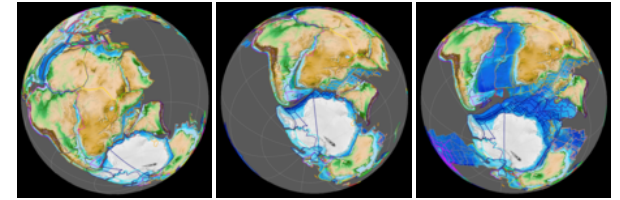
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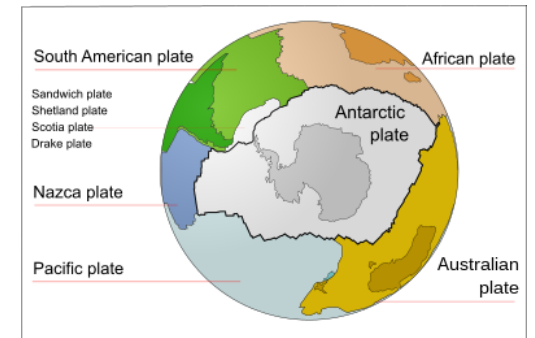


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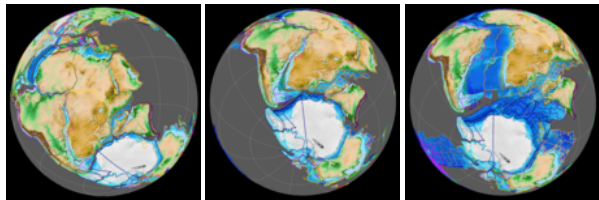
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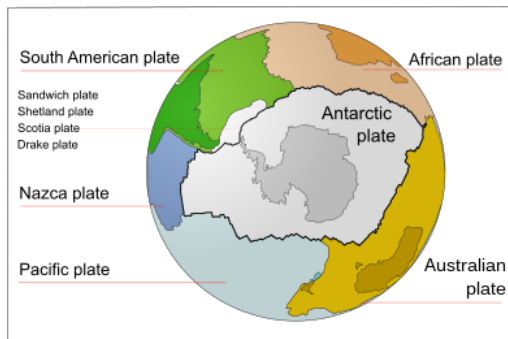
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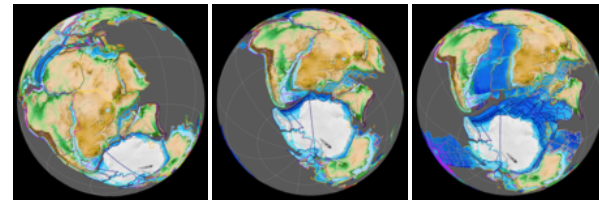
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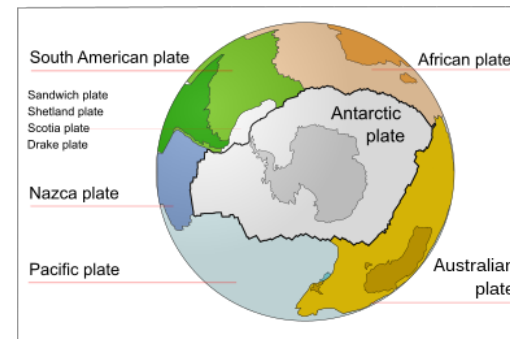
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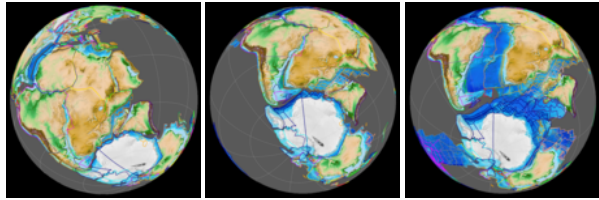
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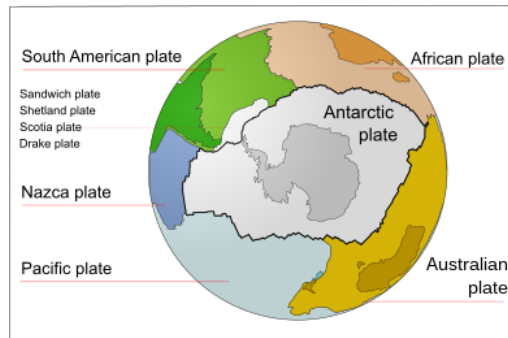


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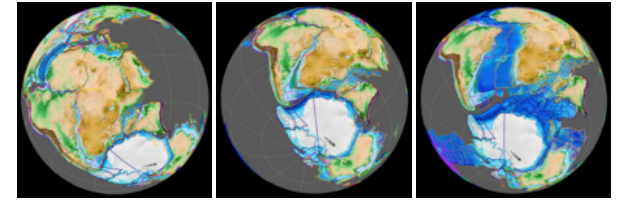
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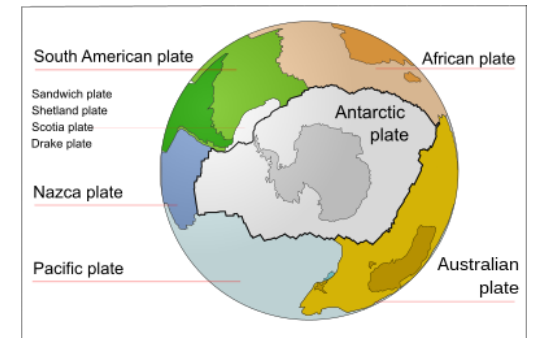


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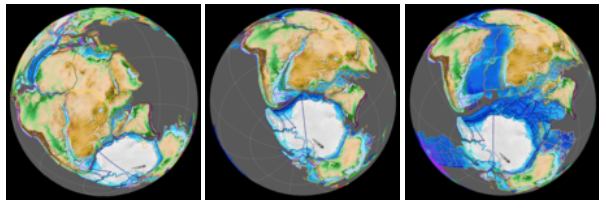
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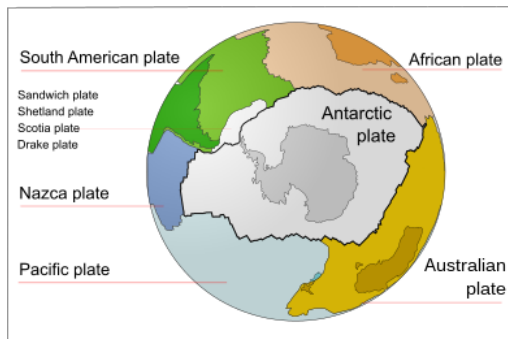
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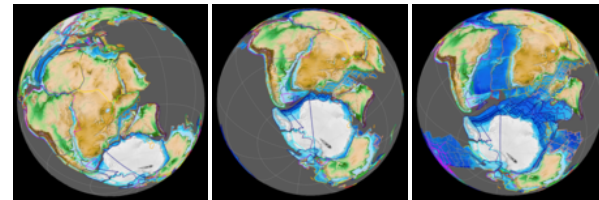
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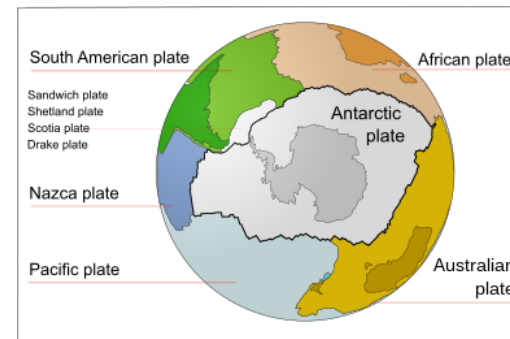
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Climate

Antarctica is the coldest, windiest, and driest of Earth's continents.^[1] The lowest natural air temperature ever recorded on Earth was −89.2 °C (−128.6 °F) at the Russian Vostok Station in Antarctica on 21 July 1983.^[68] A lower air temperature of −94.7 °C (−138.5 °F) was recorded in 2010 by satellite—however, it may have been influenced by ground temperatures and was not recorded at a height of 2 m (7 ft) above the surface as required for official air temperature records.^[69] Average temperatures can reach a minimum of between −80 °C (−112 °F) in the interior of the continent during winter and a maximum of over 10 °C (50 °F) near the coast in summer.^[70]



Antarctica is a polar desert with little precipitation; the continent receives an average equivalent to about 150 mm (6 in) of water per year, mostly in the form of snow. The interior is dryer and receives less than 50 mm (2 in) per year, whereas the coastal regions typically receive more than 200 mm (8 in).^[71] In a few blue-ice areas, the wind and sublimation remove more snow than is accumulated by precipitation.^[72] In the dry valleys, the same effect occurs over a rock base, leading to a barren and desiccated landscape.^[73] Antarctica is colder than the Arctic region, as much of Antarctica is over 3,000 m (9,800 ft) above sea level, where air temperatures are colder. The relative warmth of the Arctic Ocean is transferred through the Arctic sea ice and moderates temperatures in the Arctic region.^[74]

Regional differences

East Antarctica is colder than its western counterpart because of its higher elevation. Weather fronts rarely penetrate far into the continent, leaving the centre cold and dry, with moderate wind speeds. Heavy snowfalls are common on the coastal portion of Antarctica, where snowfalls of up to 1.22 m (48 in) in 48 hours have been recorded. At the continent's edge, strong katabatic winds off of the polar plateau often blow at storm force. During the summer, more solar radiation reaches the surface at the South Pole than at the equator because of the 24 hours of sunlight received there each day.^[1]

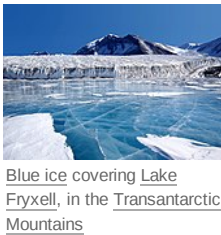
Climate change

Over the second half of the 20th century, the Antarctic Peninsula was the fastest-warming place on Earth, closely followed by West Antarctica, but temperatures rose less rapidly during the early 21st century.^[75] Conversely, the South Pole, located in East Antarctica, barely warmed during much of the 20th century, but temperatures rose three times the global average between 1990 and 2020.^[76] In February 2020, the continent recorded its highest temperature of 18.3 °C (64.9 °F), which was 0.8 °C (1.4 °F) higher than the previous record attained in March 2015.^[77]

There is some evidence that surface warming in Antarctica is due to human greenhouse gas emissions,^[78] but it is difficult to determine due to internal variability.^[79] A main component of climate variability in Antarctica is the Southern Annular Mode (a low-frequency mode of atmospheric variability of the Southern Hemisphere), which showed strengthened winds around Antarctica in the summer of the later decades of

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Glaciers and floating ice

Precipitation in Antarctica occurs in the form of snow, which accumulates and forms the giant ice sheet that covers the continent.^[81] Under the force of gravity, the ice flows towards the coast. The ice then moves into the ocean, often forming vast floating ice shelves. These shelves can melt or form icebergs that eventually disintegrate when they reach warmer ocean waters.^[82]

Sea ice and ice shelves

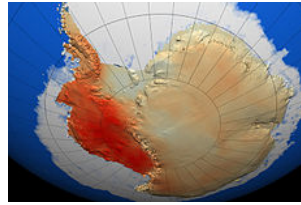
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The melting of the ice shelves does not contribute much to sea level rise, as the floating ice displaces its own mass of water, but the ice shelves act to stabilize the land ice. They are vulnerable to warming water, which has caused large ice shelves to collapse into the ocean.^[86] The loss of ice shelf "buttressing" has been identified as the major cause of ice loss on the West Antarctic ice sheet, but has also been observed around the East Antarctic ice sheet.^[87]

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Ice sheet loss and sea level rise

Antarctica contains about 90% of the world's ice. If all of this ice were melted, global sea levels would rise about 58 m (190 ft).^[93] In addition, Antarctica stores around 70% of global freshwater as ice.^[94] The continent is losing mass due to the increased flow of its glaciers toward the ocean.^[95] The loss of mass from Antarctica's ice sheets is partially offset by additional snow falling back onto it.^[96] A 2018 systematic review study estimated that ice loss across the entire continent was 43 gigatonnes (Gt) per year on average



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Temperature change per decade (degrees Celsius)
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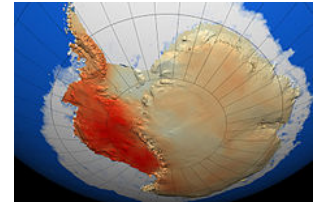
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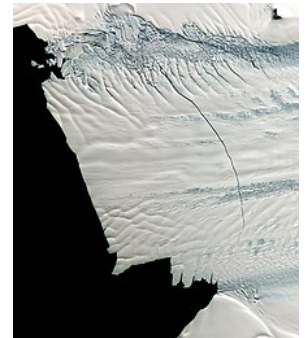
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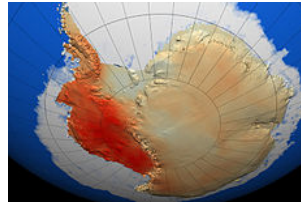
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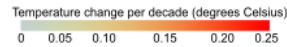
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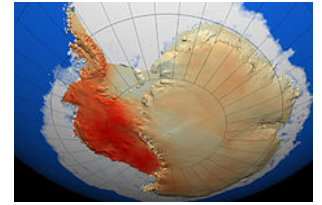
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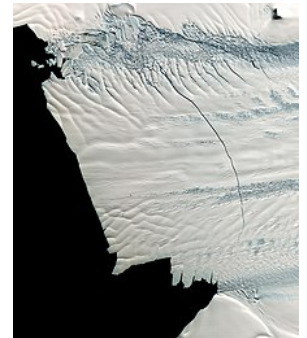
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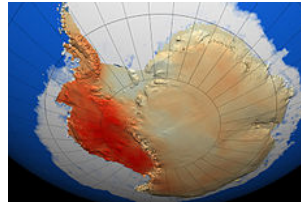
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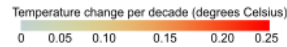
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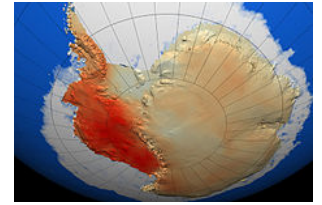
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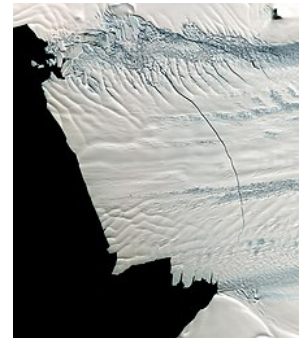
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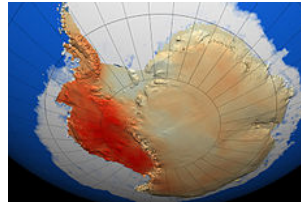
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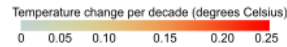
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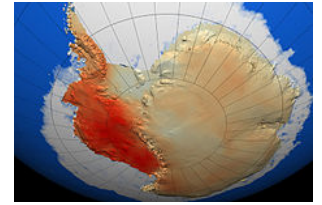
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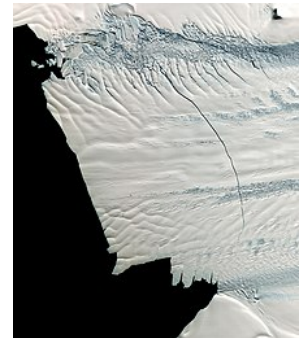
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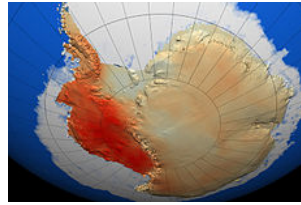
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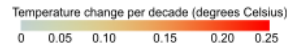
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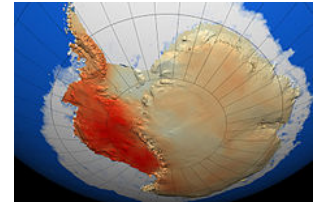
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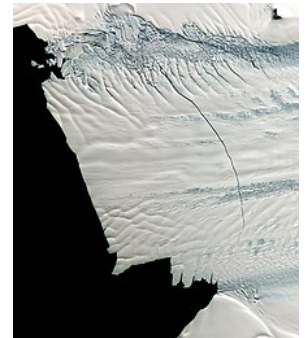
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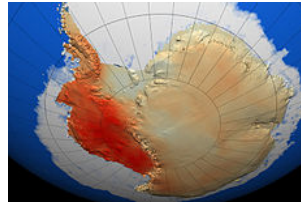
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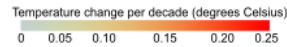
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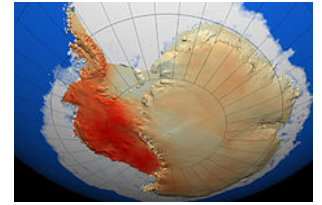
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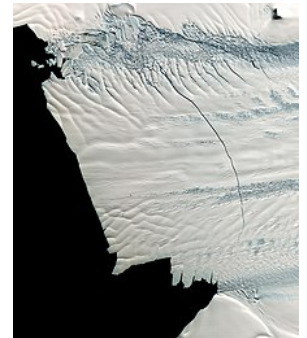
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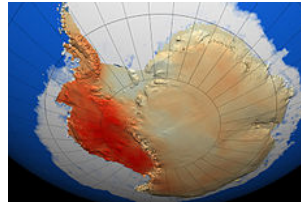
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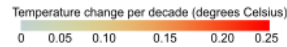
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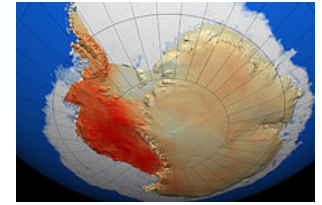
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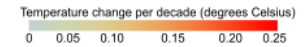
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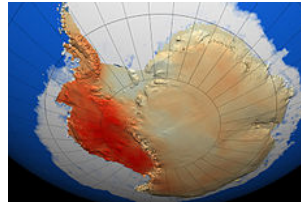
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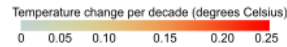
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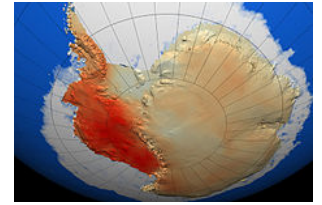
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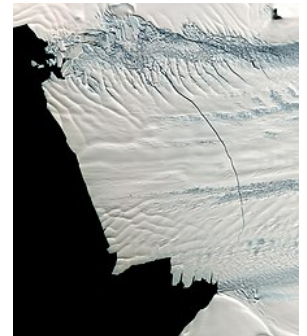
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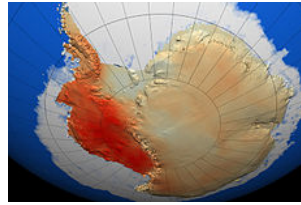
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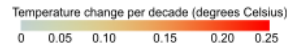
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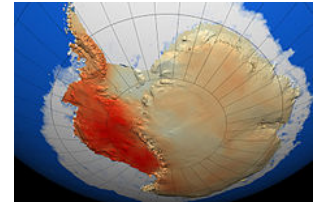
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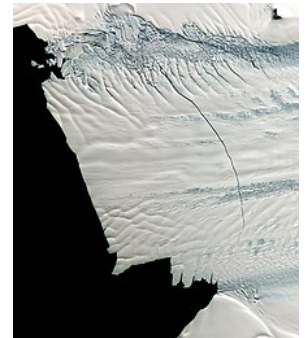
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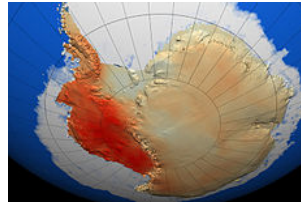
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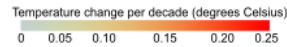
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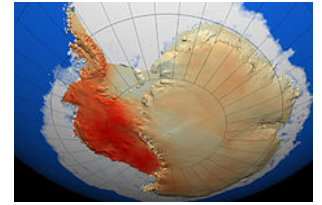
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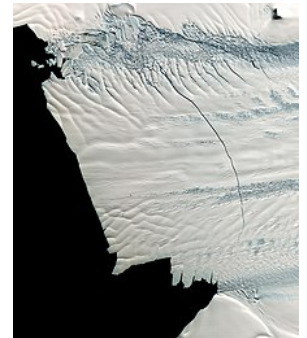
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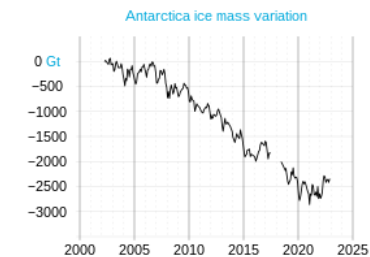
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Most of the ice loss has taken place on the Antarctic Peninsula and West Antarctica.^[98] Estimates of the mass balance of the East Antarctic Ice Sheet as a whole range from slightly positive to slightly negative.^{[95][99]} Increased ice outflow has been observed in some regions of East Antarctica, particularly at Wilkes Land.^[95]

Future projections of ice loss depend on the speed of climate change mitigation and are uncertain. Tipping

points have been identified in some regions; when a certain threshold warming is reached, these regions may start melting at a significantly faster rate. If average temperatures were to begin to fall, the ice would not immediately be restored.^[100] A tipping point for the West Antarctic ice sheet is estimated to be between 1.5 and 2.0 °C (2.7 and 3.6 °F) of global warming. A full collapse would likely not take place unless warming reaches between 2 and 3 °C (3.6 and 5.4 °F), and may occur within centuries under pessimistic assumptions. This full collapse would lead to 2 to 5 meters (6.6 to 16.4 feet) of sea level rise. At 3 °C, parts of the East Antarctic ice sheet are also projected to be fully lost, and total ice loss would lead to around 6 to 12 meters (20 to 39 feet) or more of sea level rise.^[101]

Ozone depletion

Scientists have studied the ozone layer in the atmosphere above Antarctica since the 1970s. In 1985, British scientists, working on data they had gathered at Halley Research Station on the Brunt Ice Shelf, discovered a large area of low ozone concentration over Antarctica.^{[102][103]} The 'ozone hole' covers almost the whole continent and was at its largest in September 2006;^[104] the longest-lasting event occurred in 2020.^[105] The depletion is caused by the emission of chlorofluorocarbons and halons into the atmosphere, which causes ozone to break down into other gases.^[106] The extreme cold conditions of Antarctica allow polar stratospheric clouds to form. The clouds act as catalysts for chemical reactions, which eventually lead to the destruction of ozone.^[107] The 1987 Montreal Protocol has restricted the emissions of ozone-depleting substances. The ozone hole above Antarctica is predicted to slowly disappear; by the 2060s, levels of ozone are expected to have returned to values last recorded in the 1980s.^[108]

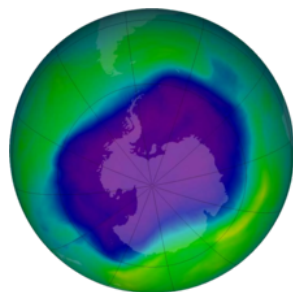
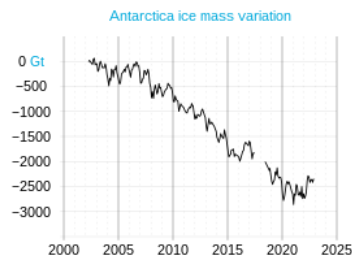


Image of the largest hole in the ozone layer recorded, in September 2006

The ozone depletion can cause a cooling of around 6 °C (11 °F) in the stratosphere. The cooling strengthens the polar vortex and so prevents the outflow of the cold air near the South Pole, which in turn cools the continental mass of the East Antarctic ice sheet. The peripheral areas of Antarctica, especially the Antarctic Peninsula, are then subjected to higher temperatures, which accelerate the melting of the ice.^[103]



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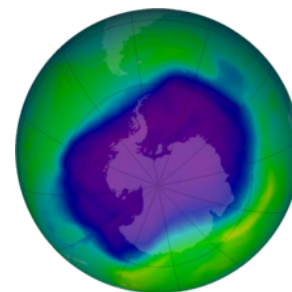
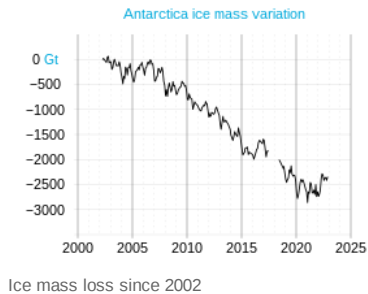


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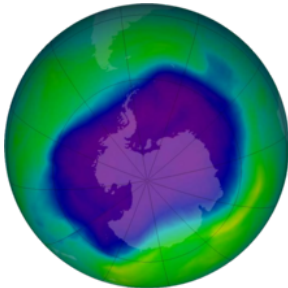
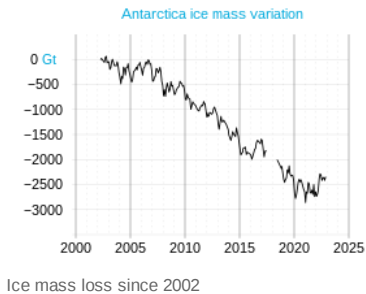


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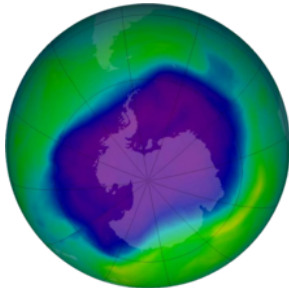
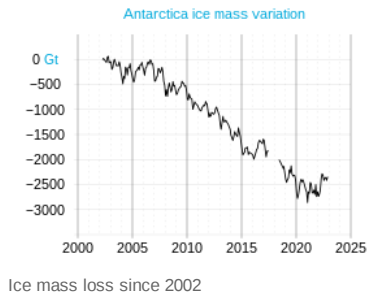


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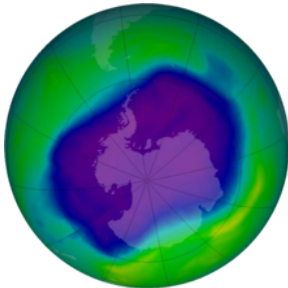
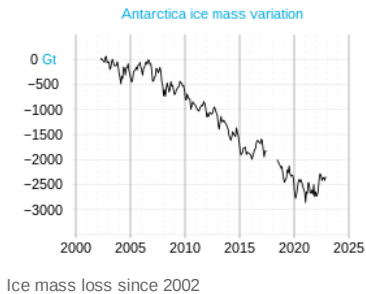


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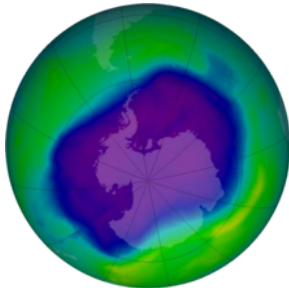
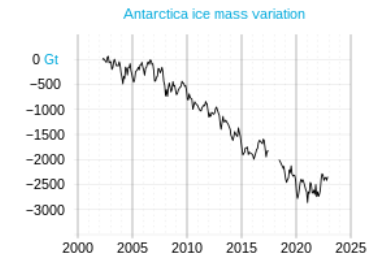


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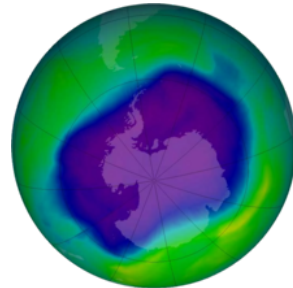
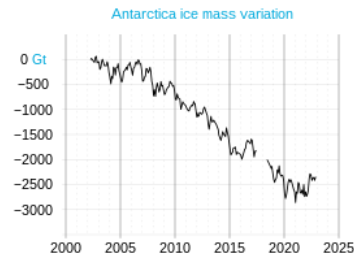


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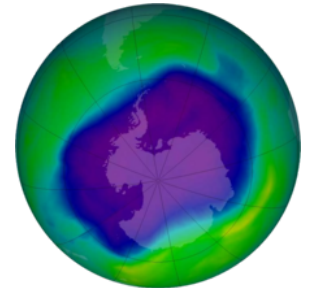
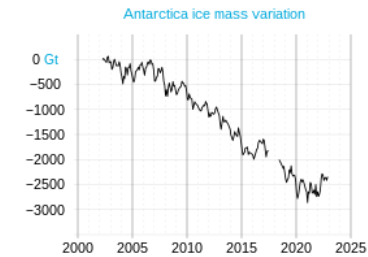


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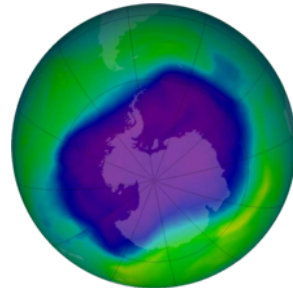
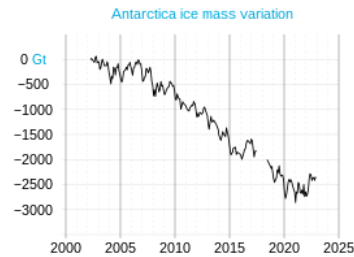


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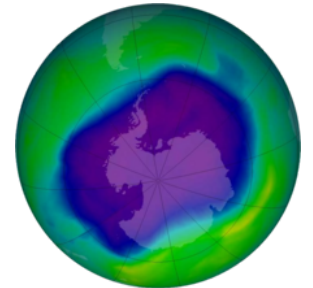
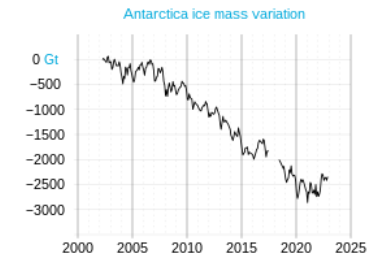


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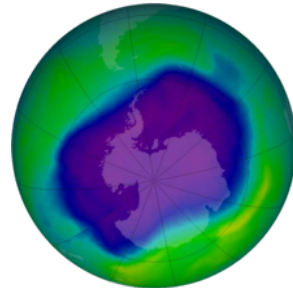
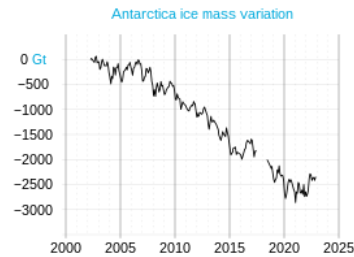


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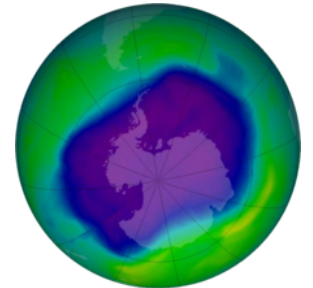
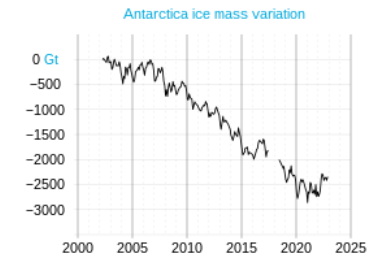


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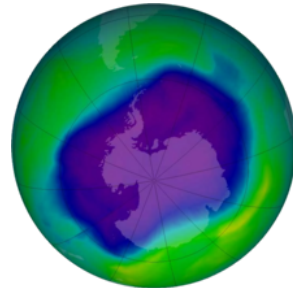
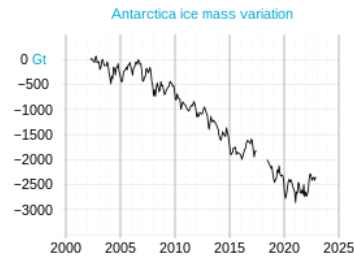


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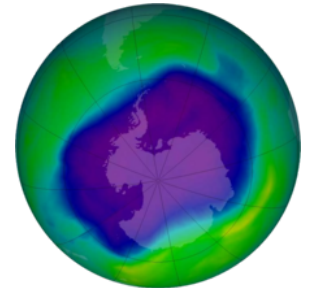
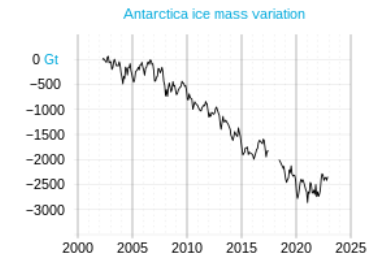


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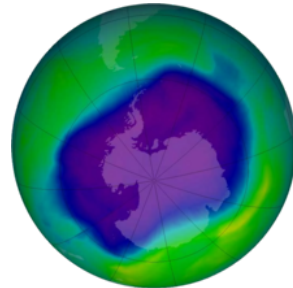
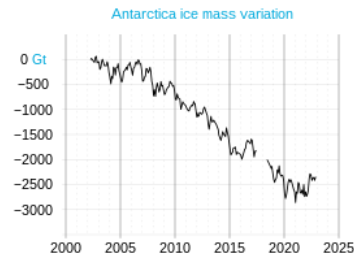


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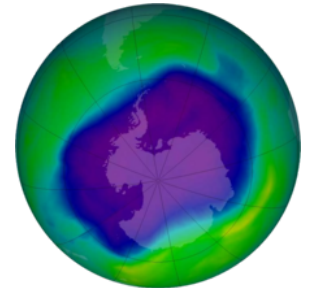
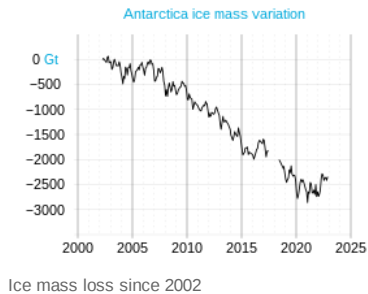


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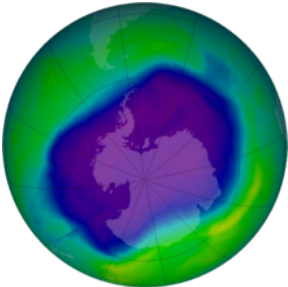
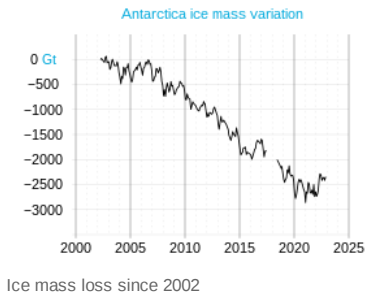


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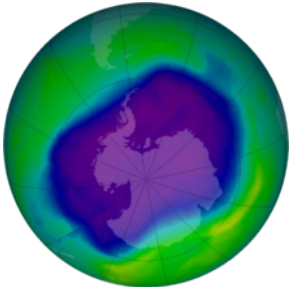
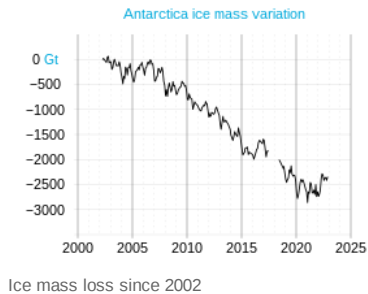


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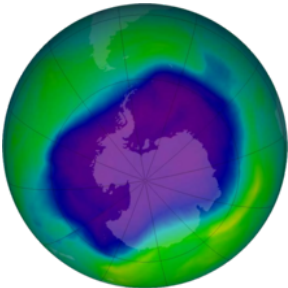
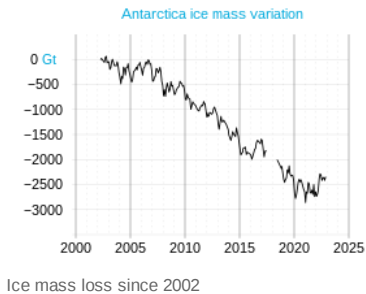


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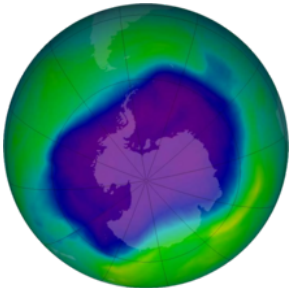


Image of the largest hole in the ozone layer recorded, in September 2006

The ozone depletion can cause a cooling of around 6 °C (11 °F) in the stratosphere. The cooling strengthens the polar vortex and so prevents the outflow of the cold air near the South Pole, which in turn cools the continental mass of the East Antarctic ice sheet. The peripheral areas of Antarctica, especially the Antarctic Peninsula, are then subjected to higher temperatures, which accelerate the melting of the ice.^[103]

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The simplified morphology of such fungi, along with their similar biological structures, metabolism systems capable of remaining active at very low temperatures, and reduced life cycles, make them well suited to such environments. Their thick-walled and strongly melanised cells make them resistant to UV radiation.^[122]

The same features can be observed in algae and cyanobacteria, suggesting that they are adaptations to the conditions prevailing in Antarctica. This has led to speculation that life on Mars might have been similar to Antarctic fungi, such as *Cryomyces antarcticus* and *Cryomyces minteri*.^[122] Some of the species of fungi, which are apparently endemic to Antarctica, live in bird dung, and have evolved so they can grow inside extremely cold dung, but can also pass through the intestines of warm-blooded animals.^{[123][124]}

Plants

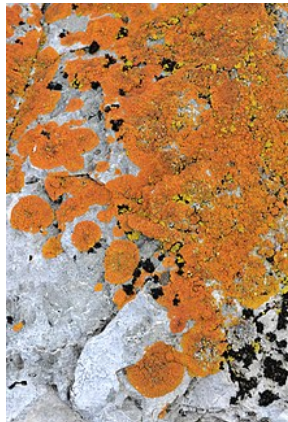
Throughout its history, Antarctica has seen a wide variety of plant life. In the Cretaceous, it was dominated by a fem-conifer ecosystem, which changed into a temperate rainforest by the end of that period. During the colder Neogene (17–2.5 Ma), a tundra ecosystem replaced the rainforests. The climate of present-day Antarctica does not allow extensive vegetation to form.^[125] A combination of freezing temperatures, poor soil quality, and a lack of moisture and sunlight inhibit plant growth, causing low species diversity and limited distribution. The flora largely consists of bryophytes (25 species of liverworts and 100 species of mosses). There are three species of flowering plants, all of which are found in the Antarctic Peninsula: *Deschampsia antarctica* (Antarctic hair grass), *Colobanthus quitensis* (Antarctic pearlwort) and the non-native *Poa annua* (annual bluegrass).^[126]

Other organisms

Of the 700 species of algae in Antarctica, around half are marine phytoplankton. Multicoloured snow algae are especially abundant in the coastal regions during the summer.^[127] Bacteria have been found as deep as 800 m (0.50 mi) under the ice.^[128] It is thought to be likely that there exists a native bacterial community within the subterranean water body of Lake Vostok.^[129] The existence of life there is thought to strengthen the argument for the possibility of life on Jupiter's moon Europa, which may have water beneath its water-ice crust.^[130] There exists a community of extremophile bacteria in the highly alkaline waters of Lake Untersee.^{[131][132]} The prevalence of highly resilient creatures in such inhospitable areas could further bolster the argument for extraterrestrial life in cold, methane-rich environments.^[133]

Conservation and environmental protection

The first international agreement to protect Antarctica's biodiversity was adopted in 1964.^[134] The overfishing of krill (an animal that plays a large role in the Antarctic ecosystem) led officials to enact regulations on fishing. The Convention for the Conservation of Antarctic Marine Living Resources, an international treaty that came into force in 1980, regulates fisheries, aiming to preserve ecological relationships.^[1] Despite these regulations, illegal fishing—particularly of the highly prized Patagonian toothfish which is marketed as Chilean sea bass in the U.S.—remains a problem.^[135]



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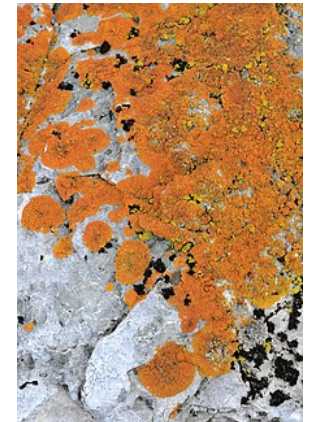
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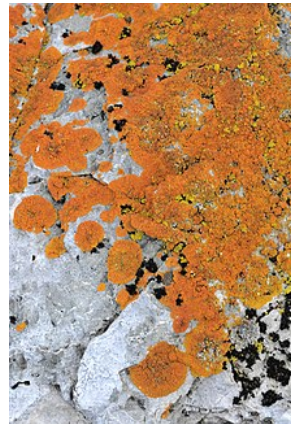
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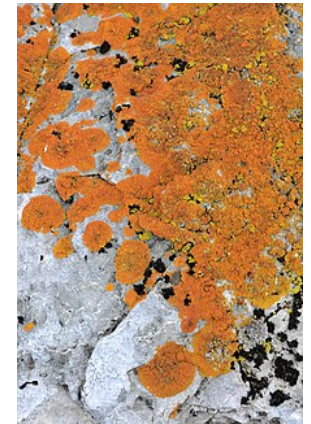
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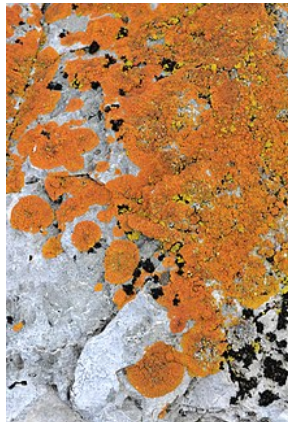
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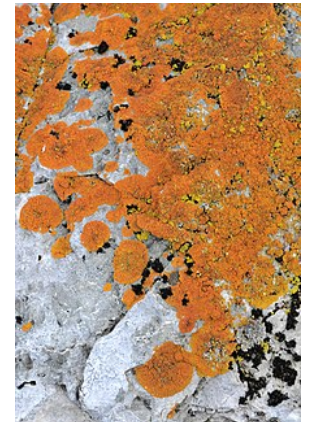
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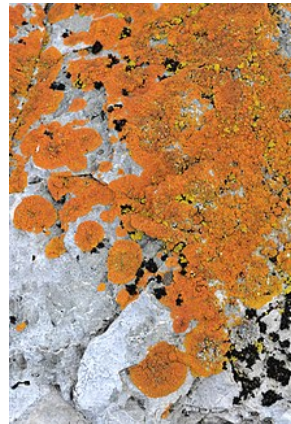
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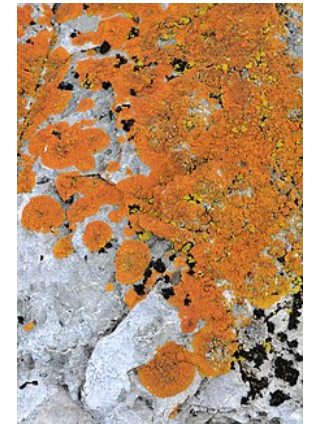
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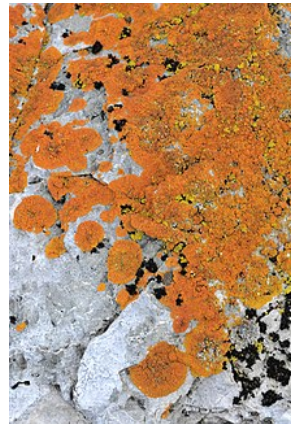
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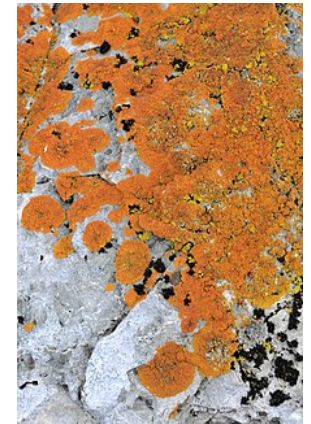
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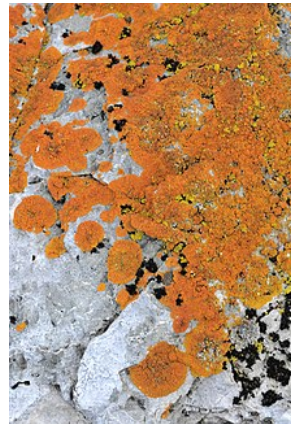
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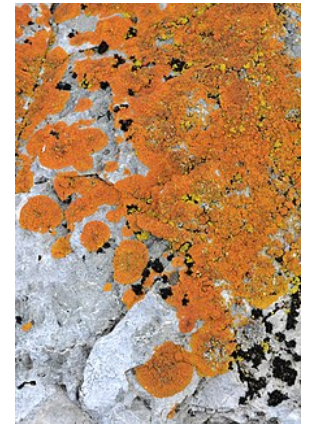
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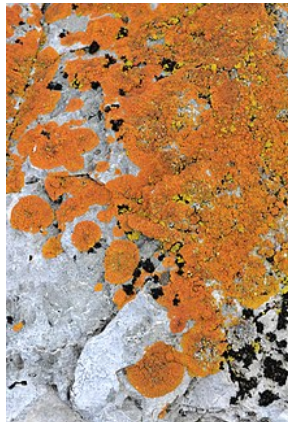
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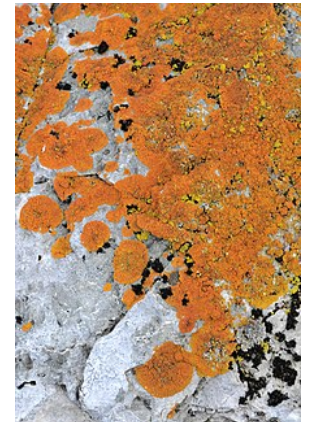
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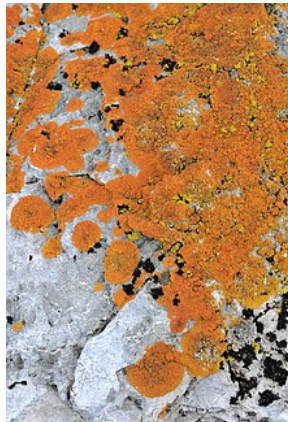
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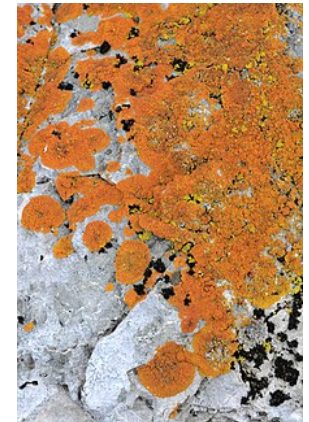
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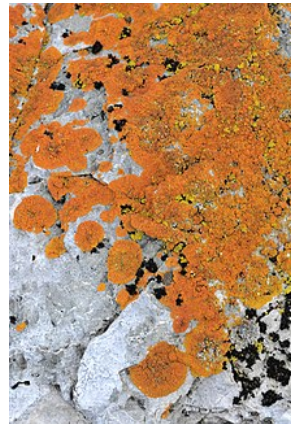
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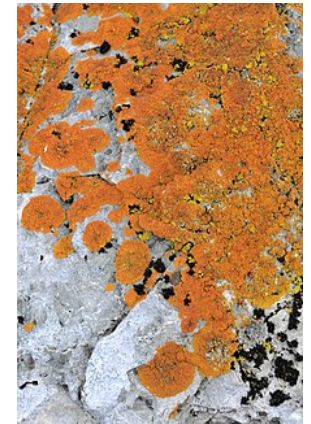
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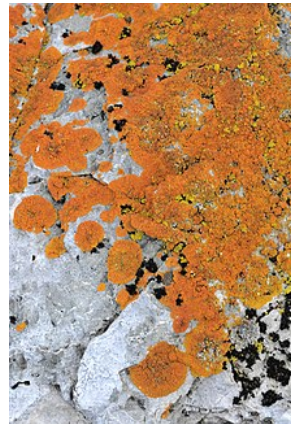
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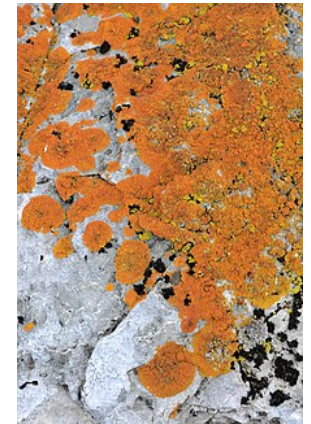
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Refuse littering the shoreline at Bellingshausen Station on King George Island, photographed in 1992



A whale in the Southern Ocean Whale Sanctuary

Ecosystems are impacted by local and global threats, notably pollution, the invasion of non-native species, and the various effects of climate change.^[134]

History of exploration

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The first semi-permanent inhabitants of regions near Antarctica (areas situated south of the Antarctic Convergence) were British and American sealers who used to spend a year or more on South Georgia, from 1786 onward. During the whaling era, which lasted until 1966, the population of the island varied from over 1,000 in the summer (over 2,000 in some years) to some 200 in the winter. Most of the whalers were Norwegian, with an increasing proportion from Britain.^{[171][note 7]}



The "ceremonial" South Pole, at Amundsen–Scott Station

The continent of Antarctica has never had a permanent resident population, although staffed research stations are continuously maintained.^[172] The number of people conducting and supporting scientific research and other work on the continent and its nearby islands varies from about 1,000 in winter to about 5,000 in the summer. Some of the research stations are staffed year-round, the winter-over personnel typically arriving from their home countries for a one-year assignment. The Russian Orthodox Holy Trinity Church at the Bellingshausen Station on King George Island opened in 2004; it is manned year-round by one or two priests, who are similarly rotated every year.^{[173][174]}

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1923	 New Zealand	 Ross Dependency	160°00'E to 150°00'W	
1931	 Norway	 Peter I Island	68°50'S 90°35'W	
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1940	 Chile	 Chilean Antarctic Territory	090°00'W to 053°00'W 80°00'W to 74°00'W claimed by the United Kingdom (1908) 74°00'W to 53°00'W claimed by the United Kingdom (1908) and Argentina (1943)	
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The cruise ship Silver Cloud in Wilhelmina Bay

Tourists have been visiting Antarctica since 1957.^[195] Tourism is subject to the provisions of the Antarctic Treaty and Environmental Protocol;^[196] the self-regulatory body for the industry is the International Association of Antarctica Tour Operators.^[197] Tourists arrive by small or medium ship at specific scenic locations with accessible concentrations of iconic wildlife.^[195] Over 74,000 tourists visited the region during the 2019/2020 season, of which 18,500 travelled on cruise ships but did not leave them to explore on land.^[198] The numbers of tourists fell rapidly after the start of the COVID-19 pandemic. Some nature conservation groups have expressed concern over the potential adverse effects caused by the influx of visitors and have called for limits on the size of visiting cruise ships and a tourism quota.^[199] The primary response by Antarctic Treaty parties has been to develop guidelines that set landing limits and closed or restricted zones on the more frequently visited sites.^[200]

Overland sightseeing flights operated out of Australia and New Zealand until the Mount Erebus disaster in 1979, when an Air New Zealand plane crashed into Mount Erebus, killing all of the 257 people on board. Qantas resumed commercial overflights to Antarctica from Australia in the mid-1990s.^[201]

Research

In 2017, there were more than 4,400 scientists undertaking research in Antarctica, a number that fell to just over 1,100 in the winter.^[1] There are over 70 permanent and seasonal research stations on the continent; the largest, United States McMurdo Station, is capable of housing more than 1,000 people.^{[202][203]} The British Antarctic Survey has five major research stations on Antarctica, one of which is completely portable. The Belgian Princess Elisabeth station is one of the most modern stations and the first to be carbon-neutral.^[204] Argentina, Australia, Chile, and Russia also have a large scientific presence on Antarctica.^[1]



An aerial view of McMurdo Station, the largest research station in Antarctica

Geologists primarily study plate tectonics, meteorites, and the breakup of Gondwana. Glaciologists study the history and dynamics of floating ice, seasonal snow, glaciers, and ice sheets. Biologists, in addition to researching wildlife, are interested in how low temperatures and the presence of humans affect adaptation and survival strategies in organisms.^[205] Biomedical scientists have made discoveries concerning the spreading of viruses and the body's response to extreme seasonal temperatures.^[206]

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Human activity

Economic activity and tourism

Deposits of coal, hydrocarbons, iron ore, platinum, copper, chromium, nickel, gold, and other minerals have been found in Antarctica, but not in large enough quantities to extract.^[193] The Protocol on Environmental Protection to the Antarctic Treaty, which came into effect in 1998 and is due to be reviewed in 2048, restricts the exploitation of Antarctic resources, including minerals.^[194]



The cruise ship Silver Cloud in Wilhelmina Bay

Tourists have been visiting Antarctica since 1957.^[195] Tourism is subject to the provisions of the Antarctic Treaty and Environmental Protocol;^[196] the self-regulatory body for the industry is the International Association of Antarctica Tour Operators.^[197] Tourists arrive by small or medium ship at specific scenic locations with accessible concentrations of iconic wildlife.^[195] Over 74,000 tourists visited the region during the 2019/2020 season, of which 18,500 travelled on cruise ships but did not leave them to explore on land.^[198] The numbers of tourists fell rapidly after the start of the COVID-19 pandemic. Some nature conservation groups have expressed concern over the potential adverse effects caused by the influx of visitors and have called for limits on the size of visiting cruise ships and a tourism quota.^[199] The primary response by Antarctic Treaty parties has been to develop guidelines that set landing limits and closed or restricted zones on the more frequently visited sites.^[200]

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An Antarctic meteorite, Allan Hills 84001 on display at the Smithsonian Museum of Natural History

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